

Sahilpreet Singh Sidhu

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EDUCATION

Chitkara University, B.Tech in Computer Science 2024 to 2028 (expected)
CGPA: 9.27 / 10 | Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, OOP

TECHNICAL SKILLS

AI and ML: LLM agents (ReAct, tool calling, MCP), RAG, embeddings, FAISS, Ollama, scikit-learn, Hugging Face Transformers, T5 fine-tuning, evaluation harnesses
Languages: Python, TypeScript, JavaScript, Rust, C++, Java, SQL
Backend and Web: FastAPI, REST, WebSockets, React, SQLite
Infra: Docker, Linux, systemd, GitHub Actions, Microsoft Azure (AI-900, AZ-900, DP-900)

PROJECTS

VirtualBuddy: local AI desktop agent | *Python, Ollama, scikit-learn, Playwright, SQLite FTS5, MCP* 2026
github.com/SahilSidhu7/VirtualBuddy

- Built a desktop agent that routes typed or spoken commands to skills (web research, scraping, file and PC tasks) with no API keys or cloud model; ships as a 29MB Windows installer.
- Replaced a per-command Ollama embedding call (about 2s) with an in-process TF-IDF + logistic-regression router (under 1ms); low-confidence commands fall through to the local LLM.
- Added a ReAct agent loop with tools, sandboxed code execution, memory, a runtime critic and MCP; fixing how requests were shaped raised a 30-task harness from 0/6 to 28/30 with no weight changes.
- Found and fixed a silent failure: a prefix match reported an uninstalled model as ready, so every LLM call 404ed unnoticed. Failures are now surfaced, and every answer is logged with a verdict.

Sentinal: local-first security monitoring (team of 4, in progress) | *Python, FastAPI, Drain3, ONNX MiniLM, Isolation Forest, Docker, React* 2026
github.com/SahilSidhu7/Sentinal

- Main contributor (34 of 48 commits). Built a CLI that builds and runs a user's app container, scans it for vulnerabilities at startup, and watches its logs for attacks with no LLM in the detection path.
- Trained the log-anomaly model (Drain3 templates, MiniLM embeddings, Isolation Forest) on real datasets including CSIC 2010, and tuned it for low false positives; 100% attack recall and 0 false positives on the 40-request demo.
- Built the hosted platform: per-project Linux environments, two browser terminals (PTY over WebSocket) and live alerts, served from one port behind admin-password auth.

2DVideoGen: closed research project, 24 logged attempts | *Python, Hugging Face Transformers (T5), custom scene DSL* 2026
github.com/SahilSidhu7/2DVideoGen

- Trained a T5 model to turn plain-English prompts into a scene-script DSL, rendered to video and scored by an automatic critic, on one 8GB laptop GPU; 100% parse/render rate on unseen prompts.
- About half of the 24 attempts failed; each is logged with its real cause. Scaling 60M to 220M parameters improved every metric but not the video, so I closed the project: capacity was not the bottleneck.

AI Website Chatbot: RAG widget | *Python, FastAPI, FAISS, Ollama, React* 2026
github.com/SahilSidhu7/ai-website-chatbot

- Built a retrieval pipeline that scrapes a website, chunks and embeds its text, and stores the vectors in a FAISS index for semantic search.
- Answers come from a local LLM grounded in the retrieved passages; the React widget embeds on any site with one script tag.

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900), Azure Fundamentals (AZ-900), Azure Data Fundamentals (DP-900)
Full Stack Open, University of Helsinki (7 ECTS, grade 5/5) | Generative AI for Everyone, DeepLearning.AI